

Linear Independence

Defintion

Defn. A collection of vectors is **linearly independent** if the only linear combination of them that equals 0 is the trivial combination (all weights zero). Otherwise it is said to be **linearly dependent**.

Note that the collection is linearly dependent if some vector in it can be written as a linear combination of the other vectors.

Key Examples

- Two vectors u and v are linearly dependent if and only if one is a multiple of the other. If they are linearly independent, then they span a plane through the origin. Further, inserting w into the collection produces a linearly independent set if and only if w is not in $\text{Span}\{u, v\}$.
- A set containing the zero vector is automatically linearly dependent.

When Homogenous System has Unique Solution?

Fact. *The columns of matrix A are linearly independent*

$\iff Ax = 0$ *has only the trivial solution*

$\iff$ *there is no free variable.*

Summary

A collection of vectors is linearly independent if the only linear combination of them that equals 0 is the trivial combination.

The homogenous system $Ax = 0$ has a unique solution precisely when the columns of matrix A are linearly independent; equivalently there is no free variable.